

Roll No.....

## MID SEMESTER EXAMINATION SEPTEMBER - 2025

Name of the Program: **B. Tech. Petroleum Engineering**

Semester: **III**

Name of the Course: **General Geology**

Course Code: **TPE 302**

Time: **90 Minutes**

Maximum Marks: **50**

**Note:**

(i) Answer all the questions by choosing any one of the sub questions.

(ii) Each question carries 10 marks.

|           |                                                                                            |            |
|-----------|--------------------------------------------------------------------------------------------|------------|
| <b>Q1</b> | <b>(10 marks)</b>                                                                          |            |
| (a)       | Define the term “geology” and outline its key areas of study.                              | CO1        |
| <b>OR</b> |                                                                                            |            |
| (b)       | Explain the different branches of geology and their specific areas of focus.               |            |
| <b>Q2</b> | <b>(10 marks)</b>                                                                          |            |
| (a)       | Describe the physical and chemical characteristics of Earth's core, mantle, and crust.     | CO1        |
| <b>OR</b> |                                                                                            |            |
| (b)       | Discuss the nebular hypothesis and its explanation of Earth's formation.                   |            |
| <b>Q3</b> | <b>(10 marks)</b>                                                                          |            |
| (a)       | Describe the methods used in the field and laboratory for mineral identification           | CO2<br>CO3 |
| <b>OR</b> |                                                                                            |            |
| (b)       | Describe the rock cycle, including the processes of weathering, erosion, and sedimentation |            |
| <b>Q4</b> | <b>(10 marks)</b>                                                                          |            |
| (a)       | What is magma? Explain its nature the processes responsible for its formation..            | CO2<br>CO3 |
| <b>OR</b> |                                                                                            |            |
| (b)       | Define texture. Discuss the various textures of igneous rocks.                             |            |
| <b>Q5</b> | <b>(10 marks)</b>                                                                          |            |
| (a)       | Define rock. Differentiate between volcanic and plutonic igneous rocks.                    | CO3        |
| <b>OR</b> |                                                                                            |            |
| (b)       | Discuss the formation and classification of igneous rocks.                                 |            |

